A Prototype Network for Hyperspectral Image Open-Set Classification Based on Feature Invariance and Weighted Pearson Distance Measurement

Yuefan Du, Xiaoping Li, *Member, IEEE*, Lei Shi, *Member, IEEE*, Fangyan Li, and Tuo Xu

***Abstract*— This study investigates the use of hyperspectral images (HSIs) in remote sensing technology, focusing on the challenges of open-set classification (OSC). The high dimensionality and complexity of HSI bring unparalleled depth and precision to remote sensing, yet pose significant classification challenges. To address these challenges, we introduce a novel prototype network based on feature invariance for open-set HSI classification (FIWPPN). This network utilizes a residual network (ResNet) architecture to extract spectral–spatial features and includes an invariance clustering module to enhance feature boundary delineation in the prototype network classification. Furthermore, we have developed a weighted Pearson distance metric to establish a measurement domain between unlabeled data and training data, facilitating open-set recognition. Experimental validation on three publicly accessible I datasets demonstrates that our method surpasses existing classification techniques in terms of classification accuracy and OSC performance.**

***Index Terms*— Feature invariance, hyperspectral images (HSIs), open-set classification (OSC), prototype network, weighted Pearson distance.**

I. Introduction

THE importance of hyperspectral images (HSIs) in remote sensing has emerged as a key research area in recent years. This advanced imaging technique captures hundreds of narrow spectral bands, providing detailed information on the reflectance or emission spectra of an object across successive bands, thus unveiling its complex chemical and physical properties [1], [2], [3]. The high dimensionality and complexity IHSI offer unprecedented depth and accuracy in remote sensing science. Analyzing the multidimensional spectral propertiIof HSI enables not only the identification and classification of surface features but also the revelation of their underlying physical, chemical, and other unique characteristics [4]. Consequently, the aInt of HSI has elevated remote sensing science to a new level. It holds wide-ranging potential applications in fields, such as environmental monitoring, agricultural management, mineral exploration, and urban planning [5], [6], [7]. However, the high dimensionality and coIexity of HSI often make traditional image classification methods inadequate for achieving satisfactory results.

Recent years have seen the advent and widespread adoption of deep learning in computer vision, includinIhe field of HSI classification. Scholars have proposed methods, such as generating adversarial networks (GANs), convolutional classification networks (CNNs), and graph neural networks (GCNsIor classifying HSI [8], [9], [10], [11]. These methods are typically applied to closed sets where both training and test data originate from the same labeling space and cover all categories. However, in practical scenarios, this assumption often does not hold true. Classification methods based on closed sets may yield incorrect results, as test data might contain new categories not included in the training data. Consequently, closed-set-based classification methods misclassify unknown samples into known categories, as illustrated in Fig. 1.

The challenge posed by new categories in testrainingot present in training data is known as the open-set identification problem. Open-set classification (OSC) is a machine learning approach where the classifier must accurately classify known classes and reject unknown classes during testing. This contrasts with the traditional "closed set" classification, which assumes that all classes are known during training. The OSC concept was first formally introduced by Scheirer et al. [12] in their 2013 paper "Toward Open Set Recognition," where they extended support vector machines (SVMs) to reject instances of classes unseen during training. In 2016, Bendale and Boult [13] developed the "OpenMax" algorithm, which modifies the probabilities generated by deep learning networks to include open space probabilities, enabling the rejection of unknown class instances. In 2019, Shu et al. [14] proposed "DOC: Deep Open Classifier," a deep learning-based solution for OSC challenges. However, to date, only a few studies have focused on OSC of HSI [15]. To address this, researchers have developed supervised and semi-supervised classification methods. Supervised techniques, such as SVMs and random

Manuscript received 17 October 2023; revised 23 November 2023 and 2 January 2024; accepted 25 January 2024. Date of publication 29 January 2024; date of current version 7 February 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62371375, Grant 62371372, Grant 62101406, and Grant 62001340; in part by the Innovation Capability Support Program of Shaanxi under Program 2022TD-37; in part by the Fundamental Research Funds for the Central Universities under Grant XJS221302 and Grant QTZX23060; and in part by the Key Chain Innovation Projects of Shaanxi under Program 2022ZDLGY01-14. *(Corresponding author: Lei Shi.)*

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Digital Object Identifier 10.1109/TGRS.2024.3359311

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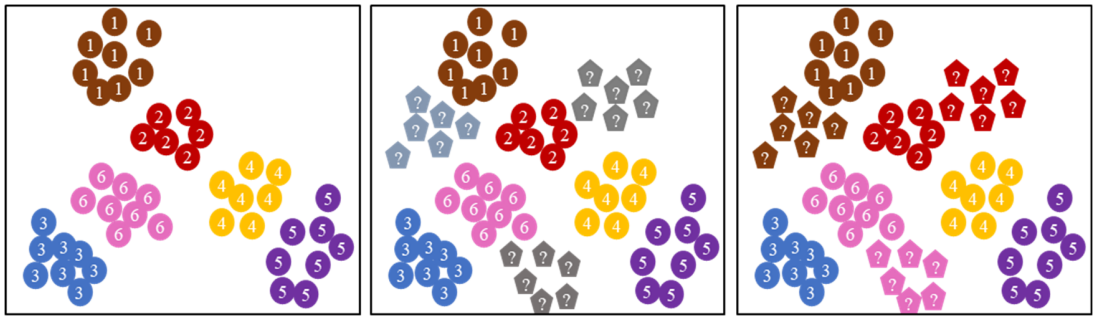

Fig. 1. Example of unknown class classification. (a) Training dataset, which contains five categories of data. (b) Test data, which contains not only the data categories present in the training data but also unknown category data. (c) Traditional closed-set classification methods misclassify the unknown category data in the test data.

forests (RFs), rely on labeled data but struggle with unknown categories in test data [16], [17]. Semi-supervised techniques, including self-training and multiview training, utilize both labeled and unlabeled data, leveraging the knowledge within unlabeled data to improve model performance, especially in OSC scenarios [18], [19]. For HSI classification, due to its complexity and diversity, there is a need for methods that are robust, flexible, and accurate for practical applications across various domains [20]. While open-set identification is becoming increasingly important for HSI classification, current methods for HSI OSC still face several key challenges.

1) *Scarcity of Training Samples:* The limited availability of training samples in OSC can predispose the model to overfitting. This overfitting risk can adversely affect classification accuracy for unknown categories.
2) *Assumptions About Distribution:* Many open-set recognition models are predicated on the assumption that the training and test sets' distributions for known categories are identical. This assumption may not always be valid in real-world scenarios, potentially impairing the model's generalization capabilities.
3) *Model Generalization Challenges:* In contrast to traditional classification, which focuses on known categories, OSC demands the model's adeptness in identifying and classifying unseen classes. This requirement for robust generalization to unknown categories is a formidable challenge, often aggravated by the scarcity of representative training data. As a result, there is an increased risk of overfitting to known classes, which could undermine the model's efficacy in accurately recognizing and classifying new categories.
4) *Threshold Determination:* A distinctive challenge in OSC is the crucial task of threshold determination. This task involves setting a boundary to differentiate known from unknown classes. Unlike traditional models, which assume that all test samples belong to known categories, open-set models face the risk of misclassifying unknown samples as known or vice versa due to inappropriate threshold settings. This task necessitates meticulous calibration and consideration of the model's prediction confidence, influenced by the training and test sets' assumed distributions.
5) *Need for Specialized Performance Metrics:* While accuracy, precision, and recall are standard metrics in traditional classification, they are insufficient for OSC due to the inclusion of unknown categories. Consequently, there is a need for specialized performance metrics that accurately assess the model's ability to differentiate between known and unknown classes. These metrics offer a more comprehensive understanding of performance by considering the correct identification of both known and unknown samples, essential for a thorough evaluation of an OSC model.

To overcome these challenges, researchers must continue to develop more robust, flexible, and accurate methods for HSI OSC.

In this article, we introduce a novel prototype network for OSC based on feature invariance and a weighted Pearson distance measurement (FIWPPN). This network involves three principal steps. First, we utilize the residual network (ResNet) to extract spectral–spatial features, employing an invariant clustering module for more robust feature boundary delineation of each class. Next, scalable prototype sets are created using discriminative features. Finally, we design a weighted Pearson distance metric for open-set identification. Experimental results on three public datasets demonstrate that our proposed method significantly surpasses other advanced classification methods. The key contributions of this study are summarized as follows.

1) In this article, we introduce a prototype network for HSI classification, centered on feature invariance, specifically exploring prototype learning for HSI open-set recognition.
2) We integrate an invariant feature module into the network to reduce intraclass distances while increasing interclass distances, thereby strengthening class boundaries and significantly enhancing the accuracy of closed-set classification results.
3) Furthermore, we propose a weighted Pearson distance metric designed for the identification of unknown samples in open-set scenarios. This approach effectively utilizes domain generalization and knowledge distillation to accurately classify unknown classes in remote sensing data.

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The structure of this article is as follows. Section II reviews the related work. Section III details the proposed FIWPPN model. Section IV presents the experiments conducted using FIWPPN on three HSI datasets, along with the results. Finally, Section V offers concluding remarks and directions for future research.

II. Related Work

In this section, we explore three facets of HSI classification: traditional methods, the application of deep learning, and open-set identification in HSI.

A. Traditional HSI Classification Methods

Traditional HSI classification approaches are grounded in both statistical and machine learning techniques.

Statistical-based methods, such as the minimum distance classifier (MDC), spectral angle mapper (SAM), linear discriminant analysis (LDA), and Gaussian mixture model (GMM), primarily depend on the statistical properties and distributions of data for classification [21], [22], [23], [24]. Machine learning-based methods, in contrast, automatically learn and extract relevant features from data, adapting more effectively to it. As a result, these methods are particularly well-suited for handling the high-dimensional nature of HSI.

SVM-based classification, a machine learning approach for classification and regression analysis, was introduced in the literature [25]. The RF method, an ensemble learning approach, was proposed in 2001 and is detailed in [26]. These methods have been extensively applied in the remote sensing image feature classification. However, their effectiveness is somewhat limited in practical HSI applications due to their inability to fully utilize multiband hyperspectral features and their tendency to introduce noise in the classification process.

To address these limitations, spectral–spatial classification methods have increasingly been adopted for HSI classification [27], [28], [29]. The local structure regularized low-rank representation (LSLRR) model for HSI classification, proposed in the literature [30], leverages local pixel similarity by integrating spatial and spectral features. In addition, Wang et al. [31] introduce a novel approach for significant band selection in HSI, enhancing significance detection. While these methods effectively adapt classification techniques to HSI, they can lead to significant misclassification in cases of insufficient training data.

B. Deep Learning Methods for HSI Classification

Deep learning, with its unique hierarchical network structure, excels in extracting informational features from data, demonstrating exceptional performance in the HSI classification. This contrasts with traditional methods, marking a significant advancement in the field [32], [33], [34].

CNNs have achieved remarkable success in the HSI classification by automatically learning and extracting high-dimensional features from HSI data, thus ensuring accurate classification results [35]. A stacked autoencoder (SAE)-based method for HSI classification is proposed in the literature [36], focusing on spatially dominant information to attain high classification accuracy. In addition, a deep belief network (DBN-based) method, utilizing a multivariate optical sensor and a constrained Boltzmann machine, is detailed in the literature [37]. Recurrent neural networks (RNNs) are also widely applied in the HSI classification [38], with literature [39] introducing an RNN classifier based on simple recurrent units that offer scalability and efficiency. While these methods typically require a substantial number of training samples, there have been developments to address this limitation. Song et al. [40] and Paoletti et al. [41] have integrated residual learning (RL) into classification networks, increasing the number of model layers while effectively mitigating overfitting. Furthermore, Wang et al. [42] introduce the use of a dense network (DenseNet) model for HSI classification, which enhances accuracy and reduces training time through feature reuse and information transfer. Despite these improvements, these methods are still limited in their applicability to OSC scenarios.

C. HSI Open-Set Classification Method

In HSI classification, the emergence of new feature categories has revealed that traditional supervised learning methods often struggle with such challenges, necessitating a method capable of handling unknown categories, namely, OSC.

The concept of zero sample learning (ZSL) was introduced in literature [43], prompting numerous subsequent studies in this field [44], [45], [46]. Although ZSL aligns more closely with practical application scenarios than previous closed-environment classification settings, accessing auxiliary information can be challenging. Dong et al. [47] suggest a domain partitioning algorithm that categorizes test samples into known, unknown, and uncertain domains, initially classifying them using probabilistic statistics and then employing the Kolmogorov–Smirnov (K–S) test for open-set identification. Another study [13] proposes a deep neural network tailored for OSC, which estimates the probability of unknown classes using an OpenMax layer. However, this approach is vulnerable to more complex generative adversarial techniques. The convolutional prototype network (CPN), proposed in literature [48], employs CNNs to learn data features and replaces the CNN's SoftMax layer with a prototype model, thus facilitating location category discrimination. Recent deep learning-based HSI classification studies, including the attention multihop graph with multiscale convolutional fusion network (AMGCFN) [49], structure preserving hypergraph convolutional network (SPHGCN) [50], dual viewpoint spectral with global spatial feature fusion network (DSGSF) [51], and rotationally invariant attention network (RIAN) [52], each contribute uniquely to HSI classification but are computationally demanding. Huang et al. [53] introduce the two-branch deep graph convolutional network (TBDGCN), capable of extracting both super pixel- and pixel-level features for HSI classification. In addition, Liu et al. [54] present a refined prototype contrast learning network (RPCL-FSL), combining supervised contrast learning (CL) and few-sample learning

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(FSL) to offer a novel perspective for small-sample HSI classification. These methods, however, may encounter issues, such as overfitting and high computational costs. Finally, Yu et al. [55] propose a two-branch attention adversarial domain adaptation (TAADA) network for cross-domain classification of HSIs, enhancing cross-domain feature representation and classification capability through adversarial learning mechanisms. Nonetheless, this method faces challenges like unstable training in adversarial learning scenarios.

While these methodologies contribute to OSC, they often struggle with overfitting and model generalization in the context of HSI classification. Therefore, this article integrates these challenges with the deep features of HSI to propose a prototype network for open-set HSI classification, based on feature invariance.

III. Method

This section provides a detailed overview of the FIWPPN model. We sequentially discuss the general structure of FIWPPN, including its feature extraction and OSC modules for HSI data.

A. FIWPPN Framework

The FIWPPN model we have developed functions as a prototype network for open-set recognition in HSI. It comprises four main components: feature extraction, prototype construction, edge reinforcement, and OSC, each crucial to the system's overall functionality.

In the feature extraction stage, a shared feature space is learned, representing each class and significantly enhancing the model's ability to identify unknown categories. The aims are multiple: to improve model generalization by sharing features across classes, increase efficiency by reducing computational load, and enhance prediction stability by providing more context for new class identification.

Subsequently, prototype construction establishes a highly representative and interpretative framework in the feature space for known classes, aiding in the differentiation between known and unknown classes. A newly encountered sample is then evaluated based on its proximity to these constructed prototypes, determining its classification as known or unknown.

The edge reinforcement process further strengthens the model's distinctive capabilities and generalization. Specifically, reinforcing the edges of known classes aids the model in better identifying unknown inputs, as these are more likely to be distant from the edges. This results in the compression of known classes and enhanced generalization to unknown categories. Through Mixup, the distribution of known samples is altered, leading to variations in the prototype model and necessitating prototype updates.

Crucially, open-set classification relies on a similarity metric domain for effective recognition. Unseen unknown class samples may initially be misclassified into known classes due to their absence during training. The model employs a method based on measures of distance or similarity to compare new samples with known classes. This comparison aids in reclassifying samples based on their proximity or dissimilarity to known classes. If a sample is distant or dissimilar, it may indicate that it belongs to an unknown class, even if initially classified as known. This mechanism thus adeptly addresses potential misclassifications, enhancing open-set recognition performance.

In summary, the FIWPPN model, through its integrated stages of operation, offers a robust solution for open-set recognition, ensuring accuracy in classifications and addressing the challenges of unknown categories.

The overall structure of the FIWPPN, as proposed in this article, is illustrated in Fig. 2.

Initially, FIWPPN extracts high-dimensional spectral features from HSI data using a ResNet network. It takes an $11 \times 11 \times B$ (where B represents the number of spectral channels in HSI data) sample block from the HSI data and inputs it into FIWPPN. The center sample label is used as the label for the entire sample block. We then evaluated various feature representations in the HSI data to construct invariant features. These features serve as the criteria for the closed-set classifier and prototype set, enhancing within-class feature consistency and interclass feature differentiation. The ResNet network is utilized again for feature extraction, and a weighted Pearson distance metric is applied to separate potential features of dissimilar samples. Network parameter updates are achieved using a cross-entropy loss and contrast loss. Finally, we complete the open-set classification of HSI data through the established closed-set classifier and prototype set, in conjunction with the open-set module.

The loss functions used in this study are as follows:

$$L_{\text{ce}} = -\frac{1}{N}\sum_{i}\sum_{c=1}^{|C|} y_{ic}\log\left(\widehat{y}_{ic}\right) \tag{1}$$

where N is a loss constant, typically $N = |C|$; $|C|$ denotes the number of categories; y_{ic} is a sign function (0 or 1), if the true category of sample i is c, then y_{ic} takes 1 at this point, and otherwise, it takes 0; and $\widehat{y}_{ic}$ is the predicted probability of observing that sample i belongs to category c.

Given that the test data for open-set classification include unknown classes not present in the training data, we update the network parameters by learning from the training data and assess the FIWPPN open-set classification performance using the test data.

B. Feature Extraction Module

FIWPPN employs the ResNet network to extract multidimensional spectral features from HSI data. As depicted in Fig. 2, the feature extraction module utilizes a 1×1 convolution kernel to extract the spectral features from HSI data and applies a filter with a stride of 64 to process these features. In addition, a maximum pooling layer is positioned before the second and third convolutional layers of the ResNet network, enhancing the data sensing range and thus capturing the spatial distribution characteristics among neighboring pixels. The network also includes batch normalization layers and rectified linear unit (ReLU) activations between the convolutional and maximum pooling layers to increase its nonlinear

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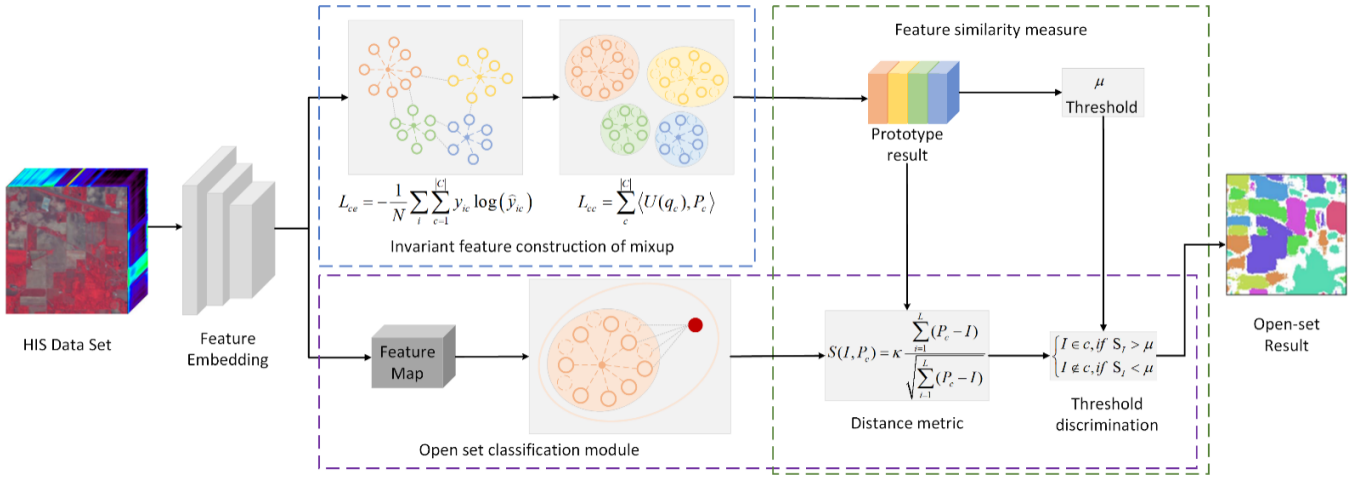

Fig. 2. Overall structure of FIWPPN.

representation capabilities, allowing for the learning of more complex patterns [56].

The spatial features of the convolutional layer are computed using the formula

$$\boldsymbol{F}_l = \delta(\boldsymbol{F}_{l-1} * w_l + k_l), \quad l \in \{1, 2, 3\} \tag{2}$$

where $\boldsymbol{F}_l$ denotes the features of the lth convolutional layer, $w_l \in R^{1\times 1}$ is the convolutional kernel, k_l is the bias, and δ is the ReLU activation function.

C. Prototyping Module

Enhancing feature representation within the same class and increasing the feature distance between different classes are crucial for ensuring FIWPPN's ability to accurately classify unknown classes. Consequently, we have incorporated a prototype building module into FIWPPN. This module evaluates the sample distribution topology by calculating the correlation between pixel data.

The modules are outlined in detail next.

$\mathbf{C} = \{\mathbf{1}, \mathbf{2}, \mathbf{L}, \mathbf{c}\}_{\mathbf{c=1}}^{|\mathbf{C}|} \subset N^+$ is presented as a collection of predefined classes, where each class is a closed set. In this framework, $|C|$ signifies the total number of known categories. Each category within this framework, denoted as c, is symbolized as $C_{,\text{where}}$ N^+ is a positive integer. When dealing with open-set recognition, labels for unidentified instances are assigned a value of $C + 1$ to distinguish them from known categories. We manage a fixed queue, exemplified by $q_{ci} = \{F_{ci}^1, F_{ci}^2, \ldots, F_{ci}^L\}$, for each distinct instance corresponding to the known class q_{ci}. This structure is designed to store the features corresponding to the ith instance q_{ci} within each known class c. F_i^L represents the Lth feature of instance q_{ci} of the cth class. For the purposes of this study, we decided to set the length of the queue, $L = 20$ to maintain a manageable and efficient system of sorting and storing instances along with their associated features.

Subsequently, by leveraging aligned second-order statistics with the correlation alignment (CORAL) and calculating the global correlation of all samples using the Jaccard index, we establish mutual invariant features among different classes of data

$$\left\langle \mathbf{q_{ci}}, \tilde{\mathbf{q}}_{\mathbf{ci}} \right\rangle = \tau \left(1 - \frac{q_{ci} \cap \tilde{q}_{ci}}{\tilde{q}_{ci} \cup q_{ci}} \right) \tag{3}$$

$$\tau = \sum_{k \neq l} \| A_k - A_l \|_F^2 \tag{4}$$

$$A_i = \frac{1}{n_k - 1} \left(X^{kT} X^k - \frac{1}{n_i} \left(1^T X^k\right)^T \left(1^T X^k\right) \right). \tag{5}$$

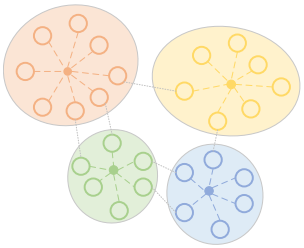

Fig. 3. Construction of relevance dendrogram.

We construct a relevance dendrogram to interconnect the data, the histogram of which is illustrated in Fig. 3.

For each instance, T relevant samples are selected, and the ratio of these T relevant samples to the same and different classes of the instance is calculated. This ratio serves as the relevant weights for prototype construction

$$\xi_{ci} = \frac{T_c}{T_{\tilde{c}}} = \frac{\{q_{ci}\}}{\{\tilde{q}_{ci}\}} \tag{6}$$

$$\boldsymbol{P}_{\mathbf{c}} = \sum \xi_{ci} \times q_{ci}. \tag{7}$$

Subsequently, we utilize the obtained prototype $\boldsymbol{P}_{\boldsymbol{c}}$ to construct the prototype set $\boldsymbol{P} = \{\boldsymbol{P}_1, \boldsymbol{P}_2, \boldsymbol{L}, \boldsymbol{P}_c\}$.

D. Boundary Reinforcement With Mixup

Mixup is a data augmentation method based on the neighborhood risk principle. This algorithm employs linear interpolation to blend two samples and their labels, thereby expanding the training data's distribution space and consequently enhancing the model's generalization ability.

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In the Mixup approach, we initially randomly select two different instances (x_i, y_i) and $(x_j, y_j)(i \neq j)$ from the same category in the training set. x_i and x_j are the two samples, with y_i and y_j as their respective labels, which are uniquely one-hot encoded. A new sample is then computed as follows:

$$\tilde{x} = \lambda x_i + (1-\lambda)x_j$$
$$\tilde{y} = \lambda y_i + (1-\lambda)y_j. \tag{8}$$

In this equation, $\lambda \in [0, 1]$ follow a Beta(α, α) distribution. The hyperparameter α controls the interpolation strength, and the larger the value, the stronger the interpolation strength. When α tends to 0, Mixup will gradually fail; $\tilde{x}$ and $\tilde{y}$ are the generated samples and labels, respectively.

In this study, we select edge data with smaller ξ-values from the same class for sample reinforcement, which in turn strengthens the class boundaries. In addition, we reinforce the class boundaries by calculating the within-class mean probability and edge probability KL divergence $\mathrm{KL}\left[U_c|\upsilon_c(y|x)\right]$. The prototype $\boldsymbol{P_c}$ is recalculated and updated according to the following equation:

$$P_{\text{new}} = \mu P + (1-\mu)P_{\text{new}} \tag{9}$$

where P_{new} denotes the updated prototype set and μ indicates the updated parameter of the prototype set. This method enables the prototype set to evolve dynamically as the data changes. Moreover, we employ the Jaccard index to calculate the contrast loss L_{cc} to further distinguish different targets in the potential space. This parameter is defined as

$$L_{\text{cc}} = \sum_{c}^{|C|} \langle U(q_c), P_{\text{new}} \rangle. \tag{10}$$

In this article, we have enhanced the training data's distribution space using Mixup. With the use of this expanded data, we recalculated the prototype $\boldsymbol{P_c}$. Here, P_{new} is used to symbolize the prototype collection after it has been updated. This process enabled the incorporation of the Jaccard coefficient in the computation of contrast loss, denoted as L_{cc}. The objective of this approach was to amplify the differentiation between various targets within the latent space. Fundamentally, the methodology involving the expansion of training data, prototype recalculation, and the application of the Jaccard Index results in a more effective separation and identification of different objectives within the system's latent space.

The total loss L_{total} of the network, comprising both the contrast loss and the classification loss, is expressed as

$$L_{\text{total}} = L_{\text{ce}} + L_{\text{cc}}. \tag{11}$$

E. Open-Set Classification

To classify unknown category samples in the test set, FIWPPN introduces an OSC module (POSM). Initially, the module employs the feature extractor to obtain features F and the predicted class index c_{pre} for each sample, excluding the closure. It then calculates the Pearson correlation coefficient between the feature embeddings and their corresponding closure class prototypes to measure the similarity metric domain S of the test set. An adaptive threshold segmentation method, based on the Otsu algorithm, is utilized to detect unknown samples in the test set.

Algorithm 1

Input: Input HSI I.

Output: The open set classification result of I.

1: According to Eq. (12), calculate the potential feature F.
2: The prediction class of input data I is obtained according to Eq. (13).
3: Calculate the similarity domain S between input data I and prototype P according to Eq. (14) and Eq. (15)
4: The open set classification result is obtained according to the threshold μ.

Return Data I open set classification result.

Specifically, when acquiring the feature embedding module f_e and the prototype set P for a known class, the test sample I is input into the feature embedding module f_e [38]

$$F = f_e(I) \tag{12}$$

where F is the extracted feature. The prediction class c_{pre} of the test sample is then obtained through the FC layer of the model [38]

$$c_{\text{pre}} = \arg\max_{c}(\text{FC}(F)). \tag{13}$$

Following these steps, we obtain the discriminative features and predicted classes of the test samples. At this stage, there is a likelihood that unknown samples will be misclassified as one of the known classes. Thus, to accurately identify unknown samples, we calculate the similarity measure domain S between the extracted features of each instance I and its prototype P_c

$$S(I, P_c) = \kappa \frac{\sum_{i=1}^{L}(P_c - I)}{\sqrt{\sum_{i=1}^{L}(P_c - I)}} \tag{14}$$

$$\kappa = \max \frac{1}{T}\sum_{i=1}^{T}\langle q_{ci}, I \rangle. \tag{15}$$

Next, we employ adaptive thresholding to categorize S into known and unknown classes, facilitating the detection of unknown samples within the similarity measure domain. This article adopts a straightforward yet effective threshold segmentation method. Specifically, we first calculate the similarity measure domain S for data of known classes to determine the threshold value μ. Then, using τ, we segregate the known and unknown classes into distinct regions, resulting in the open-set classification of instance I.

IV. Experimental Results

A. Experimental Setting

In this section, we evaluate the efficacy of FIWPPN using three HSI datasets: Pavia University, Salinas, and Indian Pines. Each of these datasets includes data samples from unknown categories [57]. Details regarding the specific data labels and the volume of data in each of these three datasets are presented in Table I.

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TABLE I
STATISTICS OF THE THREE HSI DATABASES (INCLUDING EACH CLASS NAME, UNKNOWN CLASS, AND CORRESPONDING SAMPLE NUMBER)

Pavia University			Salinas			Indian Pines		
Index	Class Name	Samples	Index	Class Name	Samples	Index	Class Name	Samples
1	Asphalt	6631	1	Brocoli-green-weeds-1	2009	1	Corn-notill	1428
2	Meadows	18649	2	Brocoli-green-weeds-2	3726	2	Corn-mintill	830
3	Gravel	2099	3	Fallow	1976	3	Grass-pasture	483
4	Trees	3064	4	Fallow-rough-plow	1394	4	Hay-windrowed	478
5	Painted-metal-sheet	1325	5	Fallow-smooth	2678	5	Soybean-notill	972
6	Bare-soil	5029	6	Stubble	3959	6	Soybean-mintil	2454
7	Bitumen	1330	7	Celery	3579	7	Soybean-clean	593
8	Self-blocking-brick	3682	8	Grapes-untrained	11271	8	Woods	1265
9	Shadows	947	9	Soil-vinyard-develop	6203	**9**	**Unknow**	**2007**
10	**Unknow**	**5163**	**10**	Corn-senesced-green-weeds	3278			
			11	Lettuce-romaine-4wk	1068			
			12	Lettuce-romaine-5wk	1927			
			13	Lettuce-romaine-6wk	916			
			14	Lettuce-romaine-7wk	1070			
			15	Vinyard-untrained	7268			
			16	Vinyard-vertical-trellis	1807			
			17	**Unknow**	**5613**			
	Total	48139		Total	58542		Total	10370

1) *Pavia University:* The dataset from Pavia University was captured by the Reflective Optical System Imaging Spectrometer (ROSIS) in the vicinity of the university. It has spatial dimensions of 610 × 340 and a spatial resolution of 1.3 m, encompassing 115 channels that span a wavelength range of 0.43–0.86 μm. Before conducting classification experiments, 12 bands affected by noise and water absorption were excluded. In addition, unlabeled buildings were identified as unknown, displaying spectral profiles markedly different from known land cover types. This dataset comprises ten groups of class table data, with the 10th group representing unknown class table data. The reference map is shown in Fig. 4.
2) *Salinas:* The Salinas dataset, sourced by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over Salinas Valley in California, USA, measures 512 × 217 in spatial dimensions and has a spatial resolution of 3.7 m. It includes 224 bands, covering a wavelength range of 0.4–2.5 μm. Before experimentation, 20 bands were removed due to water absorption. Unannotated man-made materials were classified as unknown samples. The dataset contains 17 groups of class table data, with the 17th group designated as the unknown class table data group. The reference map is provided in Fig. 5.
3) *Indian Pines:* The Indian Pines dataset, acquired by AVIRIS at the Indian Pines test site in Tippecanoe County, Northwest Indiana, USA, consists of an image with spatial dimensions of 145 × 145 and a spatial resolution of 20 m, including 220 bands that span a wavelength range of 0.43–0.86 μm. Prior to classification experiments, 20 bands were excluded due to water absorption. Following [39], classes with fewer than ten instances were discarded as tail classes and labeled unknown, resulting in eight known classes. Unlabeled roads in the dataset were also marked as unknown. This dataset comprises nine groups of class table data, with the 9th group designated as the unknown class table data group. The reference map is illustrated in Fig. 6.

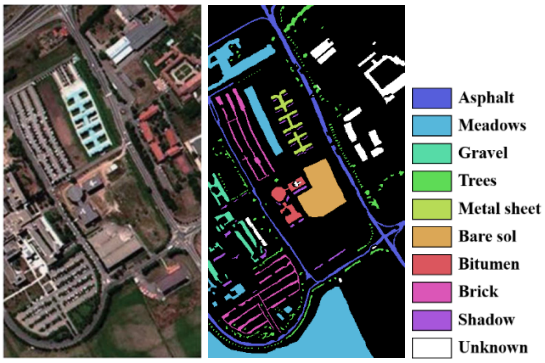

Fig. 4. Pavia University dataset. (a) Image data. (b) Ground truth. (c) Class name.

To assess the robustness and classification performance of the proposed FIWPPN method, particularly regarding its invariance properties under varying training sample sizes, we set the number of training samples for each labeled class in the experiments at 20, 50, 150, and 200. Table I details the

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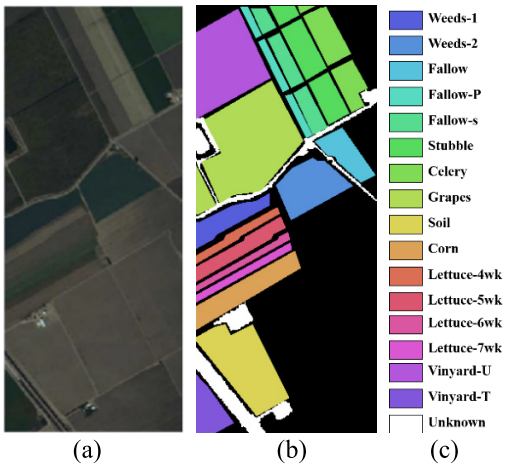

Fig. 5. Salinas dataset. (a) Image data. (b) Ground truth. (c) Class name.

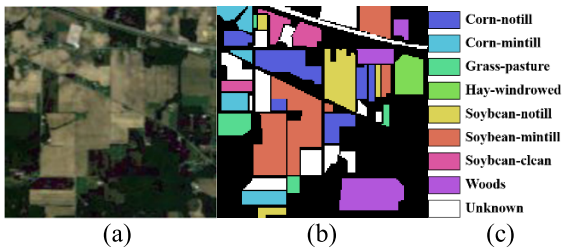

Fig. 6. Indian Pines dataset. (a) Image data. (b) Ground truth. (c) Class name.

labeled categories and the quantities of training and testing data in the three HSI datasets. Our method is evaluated using training sets of four different sizes, addressing scenarios of severe sample insufficiency, moderate insufficiency, and sample sufficiency. The remaining data from each dataset are employed as the testing set. It is important to note that during the experiments on the three datasets, none of the training samples included data labeled as "Unknown" class.

In this article, we compare the 3-D convolutional neural network (3)-DCNN) [59], spectral–spatial latent reconstruction network (SSLR) [59], double-branching multipartition mechanism network (DBMA) [61], double-branching double-attention mechanism network (DBDA) [62], the open-set hyperspectral classification method MDL4OW [57], a spectral–spatial multiple layer perceptron (MLP)-like network with reciprocal points learning (SSMLP-RPL) [63], and feature consistency-based prototype network (FCPN) [56] with the FIWPPN model proposed in this article.

The experiments were conducted using mini-batch stochastic gradient descent (SGD) with a momentum of 0.9 and employed the learning rate annealing strategy from Revgrad, where $\eta_\mu = \eta_0/(1+\alpha\mu)^\beta$ is used for tuning during the SGD, in which μ is varied from 0 to 1 over the course of the training process, $\eta_0 = 0.01$, $\alpha = 10$, and $\beta = 0.75$. An input data size of 11×11 neighborhood samples was used, and the batch size was set to 64 for all networks. In addition, we augmented the training samples by performing a 90° rotation, effectively quadrupling the training set size and enhancing sample generation. For the MDL4OW method, parameters were set as per [57]. For other comparison methods, the learning rate was set to 0.005, with linear decay every 25 epochs. Backpropagation during the training phase was executed using the Adam optimizer, with a maximum of 100 training epochs [58].

Regarding the update parameter μ of prototype set, extensive experimentation has shown that when μ is set to 0.99, the model achieves the best classification results for unknown classes in HIS data. We tested different values of μ in classifying unknown classes on three types of HIS datasets, with the results displayed in Fig. 7.

In this article, overall accuracy (OA), average accuracy (AA), and Kappa coefficient are utilized as evaluation indices to gauge the effectiveness of data classification. OA represents the overall classification accuracy across the three datasets. AA is the mean classification accuracy for each data type across these datasets. The Kappa coefficient accounts for the difference between the actual classification results of the classifier and those that would occur by chance, thus reflecting the classifier's performance. The closer these three evaluation metrics are to 100%, the more effective the classifier is deemed to be. In addition, for assessing the performance of the classifier on unknown category data in the open-set classification, we use QA, which is based solely on the OA for unknown category data.

B. HSI Classification Experiments Results

In our experiments on HIS data, we initially randomized the samples within each category, subsequently selecting 20, 50, 150, and 200 samples from each, labeled with their actual class labels, to form the training dataset. The training data excluded the label "Unknown"; therefore, while selecting the training data, we did not include data labeled as "Unknown." We then combined the remaining labeled data with the "Unknown" labeled data to construct the test dataset. These experiments evaluated the impact of the number of training samples on the classification accuracy of HIS data, particularly concerning open-set classification accuracy. To mitigate the influence of random variability and ensure the reliability of our results, we conducted five repetitions of these procedures and analyzed the algorithm's performance based on the average classification outcomes.

We compared the results of the algorithm proposed in this article with several previously mentioned algorithms across three experimental datasets. The open-set classification results for HIS data are presented in Table II. Table III details the classification accuracy for each data type in the three datasets, given a sample size of 20.

The classification results for HIS data are shown in Tables IV–VI. Meanwhile, we show in Fig. 8, and in addition, Figs. 8–10 display the performance of both the proposed method and comparative methods on the three datasets, with a sample size of 200.

As indicated in Table II, 3-DCNN, SSRN, DBMA, and DBDA exhibit subpar classification accuracy for unknown category data across the three HSI datasets, regardless of the adequacy of training data. This suggests that these methods are

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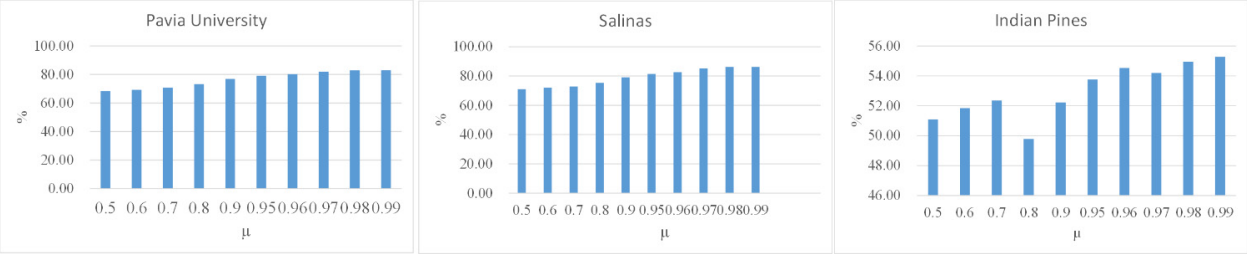

Fig. 7. Classification performance of the FIWPPN model on three datasets for unknown classes at varying values of μ.

TABLE II
OPEN-SET CLASSIFICATION RESULTS (%)

Sample size	DATASET	3-DCNN	SSRN	DBMA	DBDA	MDL4OW	SSMLP-RPL	FCPN	This paper
20	Pavia University	0.58 ±15.27	9.97±20.45	2.76±0.45	4.47±17.24	19.66±1.91	69.87±6.74	80.13±0.24	**83.04**±6.72
	Salinas	3.41±20.12	15.02±15.24	3.34±0.62	10.78±13.06	66.02±9.14	62.14±7.82	84.79±1.58	**86.33**±4.34
	Indian Pines	0.85±10.31	7.10±17.6	4.85±1.31	3.65±12.77	11.39±2.78	**74.48**±6.48	51.71±1.22	55.27±4.32
50	Pavia University	0.52±17.85	9.05±19.78	2.56±0.65	4.03±16.25	20.55±2.01	70.82±6.53	81.22±0.61	**82.40**±6.21
	Salinas	2.56±24.65	16.07±14.34	3.31±0.21	8.77±14.95	66.02±9.04	64.27±7.03	83.65±1.82	**85.97**±4.61
	Indian Pines	0.43±12.64	5.26±16.54	4.98±1.25	3.26±13.75	14.77±2.22	**76.25**±6.98	52.16±0.99	55.55±4.25
100	Pavia University	0.41±13.56	7.29±19.51	2.17±0.57	3.54±17.82	19.72±2.45	71.51±6.54	78.24±0.37	**81.34**±6.33
	Salinas	1.24±22.84	13.11±14.57	3.26±0.64	9.26±14.82	65.24±8.94	66.44±7.07	82.12±1.02	**85.37**±4.27
	Indian Pines	0.54±14.07	4.05±18.24	4.26±1.18	3.98±14.32	12.05±2.46	**77.02**±6.82	54.77±0.89	56.00±4.21
150	Pavia University	0.33±15.98	6.04±18.90	1.58±0.42	2.82±19.85	20.12±2.76	72.62±5.41	78.36±0.32	**80.27**±6.32
	Salinas	1.76±19.55	14.50±16.37	3.58±0.60	5.24±13.62	60.12±8.78	68.82±6.92	82.98±1.01	**84.77**±4.22
	Indian Pines	0.25±17.89	6.99±16.25	4.05±1.26	4.12±13.76	16.28±2.05	**78.31**±6.42	53.34±1.13	56.46±4.15
200	Pavia University	0.24±16.85	4.83±18.66	0.79±0.62	2.00±17.99	23.75±2.92	73.54±5.27	77.98±0.27	**79.21**±6.24
	Salinas	1.13±23.73	12.66±15.29	3.54±0.66	3.46±14.07	63.85±8.46	70.07±7.18	80.92±1.29	**84.17**±4.31
	Indian Pines	0.46±17.98	3.11±16.83	4.17±1.17	3.82±13.46	17.81±1.94	**79.24**±6.37	55.86±1.15	56.92±4.17

inadequate for accurate open-set classification. The MDL4OW method, which calculates unknown scores by comparing original and reconstructed data, thereby distinguishing unknown samples from known classes, demonstrates superior open-set classification capabilities. Hence, MDL4OW achieves better open-set classification accuracy. SSMLP-RPL and FCPN, which are recent open-set classification methods for HSI data, approach classification differently. SSMLP-RPL completes the open-set classification by establishing a feature learning framework based on reciprocal points learning (RPL), while FCPN enhances model discriminative ability through a class-based module, in conjunction with the prototype-guided open-set module (POSM), to effectively identify open sets in HIS data.

Consequently, these two methods exhibit higher open-set classification accuracy compared to others. In contrast, the FIWPPN method, leveraging correlation dendrograms and the weighted Pearson distance similarity measure, achieved higher accuracy in the open-set classification of unknown categories in the Pavia University and Salinas datasets than the other methods. However, FIWPPN's performance on the Indian Pines dataset was not as effective as the SSMLP-RPL method.

This disparity is attributed to the highly heterogeneous data distribution in the Indian Pines dataset, where the amount of data for different labels varies significantly, with a maximum difference of approximately 60 times. Analyzing the FIWPPN model's classification accuracy for unknown label data across various datasets and different training sample sizes in Table II, it is apparent that the size of the training sample does not directly impact the classification accuracy for unknown label data. Overall, the FIWPPN method proves effective for the open-set classification of HIS data.

As evidenced in Tables IV–VI and Figs. 8–10, the HIS open-set classification method based on invariant features

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TABLE III
CLASSIFICATION RESULTS FOR EACH CATEGORY OF THE THREE HSI DATASETS (%)

ClassNo.	3DCNN	DBMA	DBDA	MDL4OW	SSLR	FCPN	SSMLP-RPL	FIWPPN
	Pavia University dataset							
1	77.21±3.21	90.17±0.25	92.91±0.65	90.56±3.52	85.65±4.27	92.74±2.06	85.59±6.03	99.21±0.65
2	80.72±2.18	87.17±1.68	88.77±2.18	95.57±3.18	90.61±4.01	98.15±2.95	90.57+4.20	97.42±0.36
3	64.68±2.65	88.71±3.09	88.55±4.35	88.26±4.06	84.78+3.97	97.84±1.96	88.39±5.08	95.23±1.59
4	95.26±0.85	94.06±3.85	93.46±3.21	94.48±3.18	91.50+1.37	77.93±2.4	87.73±5.81	91.95±0.15
5	98.20±2.49	98.34±4.18	99.49±4.06	86.16±4.35	93.67±4.63	96.87±2.62	97.20±2.32	96.91±2.06
6	80.64±6.52	91.58±4.16	96.86±3.52	93.16±4.44	92.43±5.64	98.70±1.36	93.40±2.87	92.99±0.29
7	80.99±3.15	98.05 ±3.85	99.62±0.29	97.29±4.58	91.74±1.71	95.16±1.52	93.12±2.48	96.02±0.85
8	68.89±0.98	92.91±2.57	95.62±3.16	84.52±3.18	76.81±5.91	88.96±2.65	72.82±8.14	90.21±1.06
9	99.46±4.51	97.50±0.65	97.24±2.85	95.08±2.19	77.19±5.02	85.85±1.28	77.82±8.14	90.46±0.64
Unknown	0.58 ±15.27	2.76±0.45	4.47±17.24	19.66±1.91	52.09±11.81	80.13±0.24	69.87±6.74	83.04±6.72
ClassNo.	Salinas dataset							
1	90.16±4.56	99.80±0.95	99.90±1.59	92.58±2.59	92.53±2.49	100.00±0.25	99.45±1.02	97.65±1.56
2	99.11±0.26	99.80±5.05	99.87±3.45	95.33±0.19	98.58±0.62	99.69±2.61	98.17±1.53	98.81±0.65
3	96.24±1.85	94.31±3.48	92.00±3.49	98.78±3.14	99.04±0.98	99.22±2.48	99.60±0.25	92.63±3.15
4	96.13±1.09	98.94±2.57	97.18±1.13	99.15±2.18	82.26±1.32	99.99±2.56	89.55±0.77	91.35±1.06
5	94.95±4.06	95.94±1.64	96.88±2.15	90.42±4.65	98.48±1.40	90.59±2.19	88.70±1.03	97.5±0.36
6	98.54±2.85	99.60±1.08	99.42±2.54	96.60±1.08	97.11±2.35	89.97±3.05	94.63+3.56	95.32±2.85
7	97.31±3.65	99.46±0.64	98.93±1.98	96.00±4.75	90.42±1.09	99.97±3.21	96.33±1.23	98.32±1.65
8	69.63±0.25	68.42±3.74	66.75±5.16	75.91±0.16	76.75±9.56	80.64±2.16	87.60±2.06	90.26±1.49
9	98.02±3.69	84.32±2.85	91.78±4.18	99.62±1.49	98.96±0.51	90.06±2.85	98.34±1.12	99.36±2.00
10	93.69±3.85	95.44±2.49	97.11±0.68	93.96±2.96	95.83±1.63	91.57±2.49	94.97±2.46	96.35±1.06
11	98.61±0.26	82.29±1.05	95.17±3.05	97.57±3.18	94.07±2.10	98.35±1.65	98.48±0.65	93.24±2.06
12	89.49±4.26	94.22±0.54	99.33±1.54	99.56±2.14	99.93±0.22	94.75±1.32	90.14+0.33	96.21±2.07
13	98.64±1.59	98.33±0.68	97.88±4.35	99.45±0.65	99.53±0.40	99.25±2.64	97.79±1.65	92.16±3.02
14	98.02±3.65	98.55±5.15	99.22±1.06	99.08±2.16	81.13±0.82	96.71±1.75	96.13±1.61	90.16±1.66
15	75.60±3.25	83.37±1.68	85.81±0.25	87.53±8.25	83.25±12.44	93.28±2.18	87.28±1.99	97.21±2.19
16	97.21±0.65	93.91±3.25	98.06±6.15	97.79±4.16	65.33±5.61	95.88±2.16	74.06±3.65	95.32±1.65
ClassNo.	Indian Pines dataset							
1	36.04±2.05	50.14±3.19	70.85±3.85	77.61±3.46	76.61±1.54	85.71±2.06	88.18±3.21	84.32±0.26
2	53.98±3.24	84.25±3.46	75.38±3.46	85.58±3.82	78.75±1.99	87.01±0.52	87.75±4.61	89.25±0.84
3	76.46±2.85	60.39±2.98	90.84±3.19	85.98±0.68	83.56±0.42	69.94±1.35	81.65+3.04	72.65±0.95
4	93.80±3.18	97.95±4.06	98.95±3.68	99.21±1.08	81.20+7.45	98.78±0.68	82.60±1.77	99.36±0.52
5	44.58±2.64	79.20±3.75	81.18±1.85	82.71±2.19	77.59±0.41	75.63±2.06	84.51±3.67	82.06±0.79
6	49.12±3.85	77.77±3.46	76.75±4.09	75.55±3.82	80.29±10.71	77.05±1.08	91.62±2.11	84.32±0.65
7	55.25±3.64	68.94±3.82	86.42±3.84	78.94±4.25	72.12±0.63	95.92±1.56	94.53±1.56	98.63±0.31
8	86.73±3.39	98.35±3.59	95.12±4.28	94.77±3.08	87.61±1.54	98.36±1.20	82.31±1.23	99.25±0.35
Unknown	0.85±10.31	4.85±1.31	3.65±12.77	11.39±2.78	70.15±2.06	51.71±1.22	74.48±6.48	55.27±4.32

TABLE IV
CLASSIFICATION RESULTS FROM PAVIA UNIVERSITY (%)

Sample size		3-DCNN	DBMA	DBDA	MDL4OW	SSLR	SSMLP-RPL	FCPN	This paper
20	OA	71.74±0.84	80.68±1.79	82.65±2.21	84.98±0.72	81.79±1.75	90.73±1.27	93.11±1.72	**94.05**±0.84
	AA	74.66±0.82	84.13±0.98	85.70±2.35	84.47±1.25	83.22±2.05	88.62±1.85	91.23±1.54	**92.44**±0.85
	Kappa	65.19±0.98	76.05±2.01	78.51±2.65	81.12±0.92	84.02±3.15	87.99±1.92	91.32±1.78	**92.12**±1.06
50	OA	74.25±0.83	81.92±1.85	81.91±2.25	85.81±0.74	83.02±2.13	91.15±1.54	93.45±1.70	**94.47**±0.90
	AA	76.94±0.86	85.00±1.04	85.01±2.30	85.31±1.19	85.02±2.44	89.02±1.70	91.63±1.62	**92.70**±0.88
	Kappa	68.19±1.03	77.53±1.98	77.52±2.64	82.17±1.02	86.72±2.19	88.39±2.01	91.74±1.74	**92.48**±1.10
100	OA	78.42±0.86	83.99±1.84	83.97±2.29	87.20±0.85	85.97±2.25	92.11±1.65	94.02±1.71	**95.17**±0.82
	AA	80.73±0.82	86.45±1.06	86.48±2.34	86.72±1.06	86.31±2.64	90.69±1.66	92.28±1.53	**93.14**±0.87
	Kappa	73.19±1.05	80.00±1.99	79.98±2.61	83.91±0.94	88.64±3.34	89.81±2.13	92.43±1.72	**93.07**±1.05
150	OA	82.60±0.82	86.06±1.82	86.02±2.20	88.59±0.62	88.12±1.32	93.87±1.74	94.58±1.79	**96.49**±0.81
	AA	84.53±0.87	87.91±1.06	87.94±2.31	88.12±0.93	88.72±3.16	91.90±1.86	92.94±1.50	**93.95**±0.82
	Kappa	78.20±1.02	82.47±1.98	82.43±2.68	85.65±0.81	90.03±2.16	91.00±2.05	93.13±1.76	**94.02**±1.06
200	OA	86.78±0.89	88.13±1.72	88.08±2.25	89.98±0.69	90.17±2.06	94.32±1.65	95.15±1.75	**96.57**±0.81
	AA	88.32±0.83	89.36±0.92	89.41±2.32	89.53±1.17	91.36±2.11	92.34±1.69	93.60±1.52	**94.02**±0.83
	Kappa	83.20±1.06	84.94±1.99	84.89±2.63	87.39±0.87	92.19±3.19	92.44±1.99	93.82±1.78	**94.26**±1.00

proposed in this article generally outperforms other algorithms across the three HSI datasets. On the Indian Pines dataset, the SSMLP-RPL method, with its RPL-based learning framework, aligns more closely with the dataset's characteristics, outperforming the FIWPPN method proposed here. However, SSMLP-RPL's performance on the other two datasets is

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TABLE V
CLASSIFICATION RESULTS FROM SALINAS (%)

Sample size		3-DCNN	DBMA	DBDA	MDL4OW	SSLR	SSMLP-RPL	FCPN	This paper
20	OA	80.01±0.60	79.78±0.83	81.74±2.21	88.73±0.66	82.06±1.64	91.85±2.54	92.93±2.16	**94.05**±0.77
	AA	87.93±1.03	87.65±0.65	89.77±2.51	93.26±0.60	88.91±1.45	91.94±1.30	95.92±0.99	**96.17**±0.65
	Kappa	78.13±0.72	77.91±0.90	80.05±3.01	87.62±0.72	83.24±1.56	91.07±2.91	92.22±2.15	**93.37**±0.62
50	OA	80.59±0.65	81.25±0.82	82.98±2.25	89.29±0.61	85.41±2.64	92.69±2.55	93.21±2.07	**94.43**±0.74
	AA	88.45±1.11	88.64±0.64	90.45±2.42	93.59±0.57	89.81±1.35	93.15±1.21	96.05±1.08	**96.50**±0.66
	Kappa	78.73±0.74	79.49±0.92	81.39±2.99	88.23±0.62	86.06±1.13	91.97±2.56	92.53±2.11	**93.75**±0.68
100	OA	81.56±0.66	83.69±0.82	85.06±2.26	90.22±0.60	87.21±2.17	93.23±2.22	93.68±2.12	**95.06**±0.71
	AA	89.33±1.15	90.30±0.60	91.57±2.48	94.13±0.63	91.50±2.18	94.49±1.43	96.27±1.05	**97.04**±0.68
	Kappa	79.74±0.71	82.13±0.88	83.62±3.01	89.25±0.69	88.06±1.94	92.67±2.60	93.03±2.09	**94.37**±0.60
150	OA	82.53±0.66	86.14±0.84	87.13±2.26	91.15±0.55	92.16±3.04	94.09±2.27	94.14±2.10	**96.14**±0.72
	AA	90.20±1.06	91.95±0.60	92.70±2.51	94.67±0.59	92.80±1.43	95.18±1.33	96.49±1.06	**98.02**±0.67
	Kappa	80.74±0.80	84.77±0.90	85.85±3.12	90.27±0.66	90.15±2.06	93.39±2.83	93.54±2.13	**95.52**±0.63
200	OA	83.50±0.62	88.58±0.83	89.20±2.20	92.08±0.59	93.12±1.15	95.36±2.45	94.61±2.11	**96.33**±0.75
	AA	91.07±1.00	93.61±0.60	93.83±2.49	95.21±0.61	93.29±2.56	96.63±1.39	96.71±1.00	**98.12**±0.68
	Kappa	81.75±0.78	87.41±0.91	88.08±3.00	91.29±0.58	92.84±1.27	94.28±2.38	94.05±2.10	**95.62**±0.61

TABLE VI
CLASSIFICATION RESULTS FROM INDIAN PINES (%)

Sample size		3-DCNN	DBMA	DBDA	MDL4OW	SSLR	SSMLP-RPL	FCPN	This paper
20	OA	46.01±0.97	62.61±1.74	66.43±2.42	69.09±0.69	76.29±3.31	**86.30**±1.71	78.22±2.21	80.05±1.22
	AA	55.20±1.22	69.09±1.19	75.46±3.05	76.86±1.11	74.52±2.02	**89.50**±1.80	82.23±1.27	84.13±0.75
	Kappa	38.90±0.95	56.88±1.82	61.62±2.61	64.61±0.92	71.41±3.05	**85.08**±2.11	74.71±2.25	76.25±1.30
50	OA	49.58±1.05	65.04±1.70	68.29±2.44	71.16±0.73	77.05±2.35	**86.90**±1.72	79.31±2.10	81.16±1.29
	AA	59.53±1.56	72.35±1.25	77.68±3.13	78.68±1.14	80.89±2.59	**90.04**±1.78	83.40±1.68	85.32±0.77
	Kappa	42.91±1.19	59.67±1.84	63.70±2.74	66.94±0.85	76.52±1.04	**86.18**±2.22	75.91±2.27	77.52±1.25
100	OA	55.53±0.99	69.09±1.72	71.40±2.31	74.61±0.70	78.28±3.30	**87.61**±1.75	81.12±2.21	83.01±1.26
	AA	66.73±1.24	77.77±1.20	81.38±3.33	81.71±1.12	80.25±1.05	**90.96**±1.77	85.35±1.25	87.29±0.80
	Kappa	49.58±1.06	64.32±1.80	67.17±2.54	70.83±0.93	77.05±2.06	**87.81**±2.01	77.92±2.24	79.64±1.21
150	OA	61.49±1.00	73.15±1.76	74.50±2.40	78.06±0.74	81.40±2.57	**88.57**±1.81	82.94±2.20	85.96±1.25
	AA	73.94±1.57	83.20±1.15	85.07±3.00	84.74±1.10	84.05±2.74	**92.20**±1.78	87.29±1.37	91.00±0.77
	Kappa	56.26±1.07	68.96±1.82	70.64±2.62	74.71±0.95	78.52±3.51	**88.73**±2.19	79.93±2.12	82.57±1.24
200	OA	67.44±0.96	77.20±1.70	77.61±2.43	81.51±0.69	82.52±2.18	**89.32**±1.73	84.75±2.03	86.72±1.21
	AA	81.15±1.20	88.63±1.18	88.77±3.06	87.77±1.06	88.76±1.10	**93.34**±1.80	89.24±1.21	91.24±0.74
	Kappa	62.94±0.94	73.61±1.82	74.11±2.57	78.60±0.89	80.40±0.03	**89.84**±2.16	81.94±2.26	83.88±1.29

significantly inferior to that of FIWPPN. Analysis of the experimental results reveals that FIWPPN effectively extracts domain-invariant features from training samples, enabling the training of a closed-set classifier and prototype system with enhanced classification boundaries for HSI data. As the training sample size increases, so does FIWPPN's classification accuracy on all three datasets. Notably, when the training data size is increased from 150 to 200, the disparity in FIWPPN's performance on the OA, AA, and Kappa metrics is minimal, indicating that the proposed method achieves commendable classification results even with limited training data, a feat not matched by other algorithms. The last columns in Tables IV–VI represent the average performance of FIWPPN across all experiments. Parenthetical values denote standard deviations. This study involved five repetitions of each experiment, with observed variations recorded and the maximum variation indicated in parentheses. The stability of FIWPPN's performance is evident from the standard deviation of the experimental results.

Furthermore, Fig. 8 displays the classification results of various methods on the Pavia University dataset. The figure illustrates that the 3-DCNN method exhibits significant

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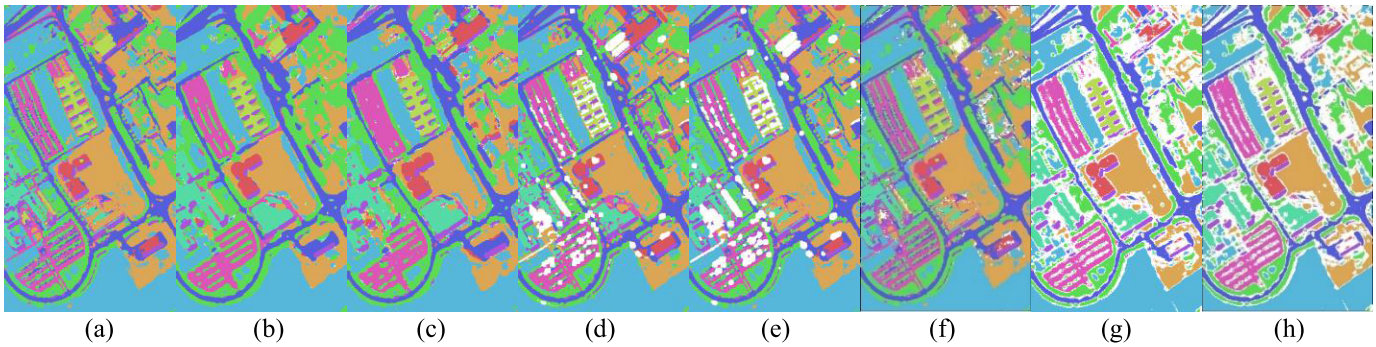

Fig. 8. Classification maps obtained by different methods on the Pavia University dataset with 200 training samples. (a) 3-DCNN, OA = 86.78%. (b) DBMA, OA = 88.13%. (c) DBDA, OA = 88.08%. (d) MDL4OW, OA = 89.98%. (e) SSRN, OA = 90.17%. (f) SSMLP-RPL, OA = 92.44%. (g) FCPN, OA = 93.82%. (h) FIWPPN (our method), OA = 96.57%.

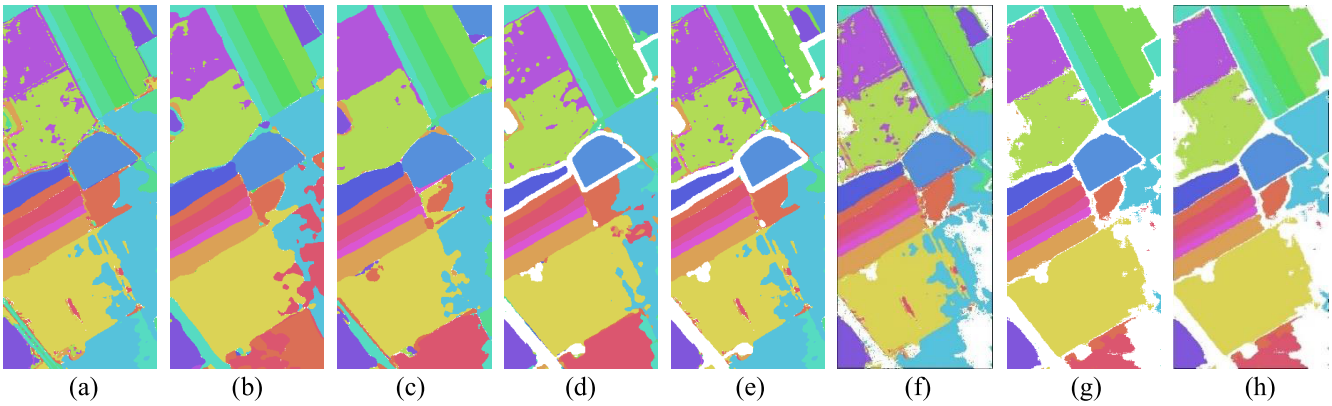

Fig. 9. Classification maps obtained by different methods on the Salinas dataset with 200 training samples. (a) 3-DCNN, OA = 83.50%. (b) DBMA, OA = 88.58%. (c) DBDA, OA = 89.20%. (d) MDL4OW, OA = 92.08%. (e) SSLR, OA = 93.12%. (f) SSMLP-RPL, OA = 95.36%. (g) FCPN, OA = 94.61%. (h) FIWPPN (Our method), OA = 96.33%.

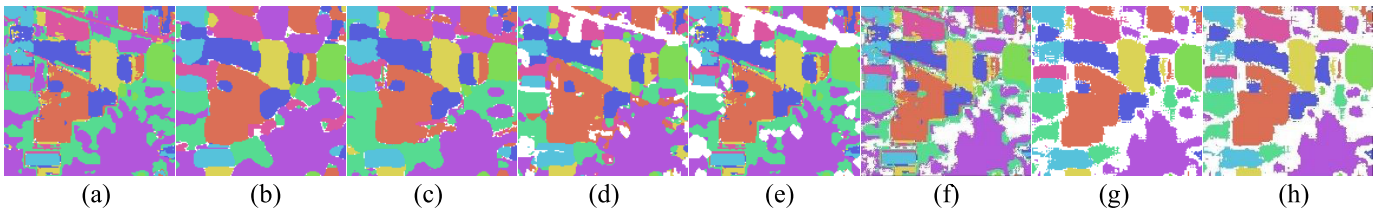

Fig. 10. Classification maps obtained by different methods on the Indian Pines dataset with 200 training samples. (a) 3-DCNN, OA = 67.44%. (b) DBMA, OA = 77.20%. (c) DBDA, OA = 77.61%. (d) MDL4OW, OA = 81.50%. (e) SSLR, OA = 82.52%. (f) SSMLP-RPL, OA = 89.32%. (g) FCPN, OA = 84.75%. (h) FIWPPN (our method), OA = 86.72%.

misclassification on this dataset. The SSLR method slightly enhances the classification performance compared to 3-DCNN. The MDL4OW method, being an open-set classification approach, yields better performance than the previously mentioned methods. The SSMLP-RPL and FCPN methods, having been proposed more recently, emphasize data and boundary augmentation in their model construction for open-set classification, resulting in improved data classification, particularly in open-set scenarios. In contrast, the method proposed in this article achieves superior classification results in most instances and effectively identifies unknown data embedded within the training dataset. This success is primarily due to the domain-invariant features and weighted Pearson distance similarity metric used in the FIWPPN method, which significantly enhances the data class boundaries of the prototype very well.

Fig. 9 presents the classification results of different methods on the Salinas dataset. The DBMA and DBDA methods, by incorporating spatial–spectral properties, show improved results on this dataset compared to the 3-DCNN method. However, their performance in recognizing unknown classes is still unsatisfactory. Fig. 10 details the classification results on the Indian Pines dataset. It is evident that the 3-DCNN, DBMA, and DBDA methods underperform in classification on this dataset. While both SSMLP-RPL and FCPN successfully identify unknown classes not included in the training data, FCPN experiences significant misclassification. The SSMLP-RPL method surpasses other algorithms on the Indian Pines dataset, but its performance on the Salinas and Pavia University datasets is less effective.

This article provides an in-depth analysis of the computational complexity of the FIWPPN method compared to others.

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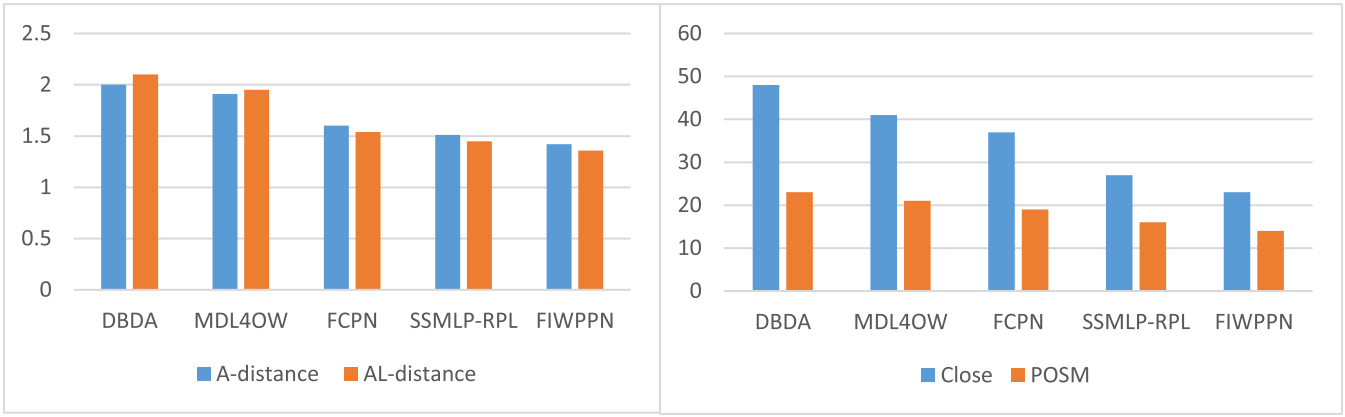

Fig. 11. (a) Classification errors of DBDA, MDL4OW, FCPN, SSMLP-RPL, and FIWPPN in Pavia University. (b) Closed-set classification and open-set classification errors of DBDA, MDL4OW, FCPN, SSMLP-RPL, and FIWPPN in Pavia University.

TABLE VII
COMPUTING TIME OF DIFFERENT APPROACHES ON THREE DATASETS WITH 20 TRAINING SAMPLES IN SECONDS

Method	PAVIA UNIVERSITY		SALINAS		INDIAN PINES	
	Train	Test	Train	Test	Train	Test
3DCNN	**31**	19	**94**	19	**45**	4
DBMA	123	45	777	43	374	8
DBDA	88	42	410	41	198	8
MDL4OW	58	31	101	25	73	7
SSLR	323	29	662	28	194	5
FCPN	256	**10**	524	**8**	142	**2**
SSMLP-RPL	300	25	701	16	196	6
FIWPPN	262	**10**	517	10	144	**2**

It documents the training and prediction times for various methods with a training sample size of 20 across the three datasets, as presented in Table VII. The FIWPPN method demonstrates similar training and testing times to the FCPN method and exhibits the fastest testing time. This efficiency is attributed to both FIWPPN and FCPN being prototype systems that use tensors for all variables when calculating the distance to the prototype, thus avoiding vector operations. FIWPPN slightly outperforms the SSMLP-RPL method in both training and testing durations. However, FIWPPN's training time is longer than that of the 3-DCNN, DBDA, DBMA, and MDL4OW methods, with 3-DCNN having the shortest training time due to its simple three-layer 3-D convolutional network structure.

These findings also have important implications. FIWPPN's efficiency in testing times suggests its suitability for real-time applications where quick response is essential. Its advantage over SSMLP-RPL in both training and testing times could make it a preferred choice in scenarios requiring both speed and accuracy. However, FIWPPN's longer training time compared to methods like 3-DCNN might be a factor in situations with large training datasets or where training time is crucial. The comparison also underscores the significance of computational optimization in machine learning methods, as evidenced by the contrasting training times of the 3-DCNN method. Ultimately, the choice of method depends on the specific application requirements, whether they prioritize speed, accuracy, or computational efficiency.

In conclusion, the FIWPPN method proposed in this article excels across all three datasets and adeptly classifies unknown categories in HIS data for open-set classification, demonstrating robust performance consistency.

C. Analysis of FIWPPN Model Performance

We employed A-distance and AL-distance to evaluate the prototype classification error. Fig. 11(a) illustrates the open-set classification errors for DBDA, MDL4OW, FCPN, SSMLP-RPL, and FIWPPN under the d_A and d_{A_L} metrics in the Pavia University task. The results indicate that both d_A and d_{A_L} are considerably smaller with FIWPPN compared to DBDA, MDL4OW, FCPN, and SSMLP-RPL, suggesting that FIWPPN not only constructs prototypes more effectively but also more efficiently differentiates between open-set points and prototypes.

Classification errors were calculated for the Pavia University dataset using DBDA, MDL4OW, FCPN, SSMLP-RPL, and FIWPPN, both without (Closs) and with open-set (POSM) modules. From Fig. 11(b), it is evident that the errors for both no-open-set (Closs) and open-set (POSM) modules are significantly lower with FIWPPN than with the other methods, further confirming FIWPPN's superiority in invariant feature extraction.

We visualized the algorithms of DBDA, MDL4OW, FCPN, SSMLP-RPL, and FIWPPN for the Pavia University dataset. Fig. 12 displays these results, where yellow points represent known samples of the prototype and purple points represent unknown samples. In Fig. 12(a) and (b), the prototypes exhibit a loose structure with inadequate capability to reject unknown classes, leading to more overlap between unknown and known classes. However, Fig. 12(e) shows that the FIWPPN algorithm, proposed in this article, aligns the prototype well with excellent cohesion, significantly rejects the unknown class, and classifies more effectively. This demonstrates that FIWPPN constructs known class prototypes more efficiently and identifies unknown classes more efficiently than DBDA, MDL4OW, FCPN, and SSMLP-RPL.

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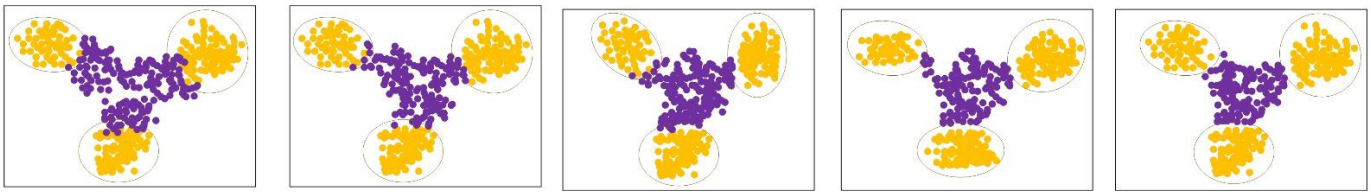

Fig. 12. Algorithmic alignment visualization [(a)–(e) denote the DBDA, MDL4OW, FCPN, SSMLP-RPL, and FIWPPN algorithm visualization results, respectively].

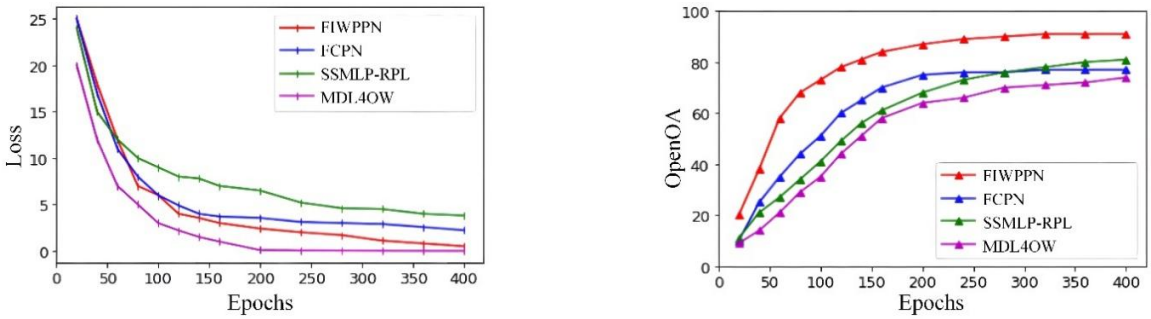

Fig. 13. Model convergence and OpenOA performance. (a) Comparison of model convergence under different loss functions. (b) Model OpenOA performance comparison.

TABLE VIII
FIWPPN ABLATION RESULTS (%)

Sample size	DATASET	FIWPPN	WPPN (FIWPPN without invariant feature)			FIPN (FIWPPN without weighted Pearson)		
			Accuracy Rate	Accuracy Variation	Mean Variation	Accuracy Rate	Accuracy Variation	Mean Variation
20	Pavia University	83.04	60.72	-22.32	-23.86	57.46	-25.58	-27.11
	Salinas	86.33	61.44	-24.89		58.54	-27.79	
	Indian Pines	55.27	30.89	-24.38		27.32	-27.95	
50	Pavia University	82.40	60.95	-21.45	-23.78	57.48	-24.92	-26.89
	Salinas	85.97	61.73	-24.24		58.73	-27.24	
	Indian Pines	55.55	29.91	-25.64		27.05	-28.5	
100	Pavia University	81.34	57.85	-23.49	-22.97	55.39	-25.95	-25.62
	Salinas	85.37	62.75	-22.62		59.43	-25.94	
	Indian Pines	56.00	33.19	-22.81		31.03	-24.97	
150	Pavia University	80.27	57.39	-22.88	-22.57	55.60	-24.67	-24.23
	Salinas	84.77	63.43	-21.34		61.65	-23.12	
	Indian Pines	56.46	33.04	-23.42		31.57	-24.89	
200	Pavia University	79.21	58.94	-20.27	-20.70	55.99	-23.22	-22.95
	Salinas	84.17	63.42	-20.75		62.01	-22.16	
	Indian Pines	56.92	35.79	-21.13		33.44	-23.48	

A comprehensive assessment of model convergence for DBDA, MDL4OW, FCPN, SSMLP-RPL, and FIWPPN was conducted, using various loss functions within the context of the Pavia University dataset. The convergence results are depicted in Fig. 13(a). In addition, we analyzed the OpenOA efficacy of these models, with comparative findings shown in Fig. 13(b). Fig. 13 indicates that while MDL4OW achieves the fastest convergence rate, its OpenOA capability is not optimal, suggesting a compromise in performance for open-set classification tasks on HSI datasets. Conversely, the FIWPPN method, advocated in our study, excels in both convergence speed and OpenOA efficiency within the same epoch range.

D. FIWPPN Model Ablation Analysis

The FIWPPN method introduced in this article incorporates domain-invariant features with a weighted Pearson distance

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similarity measure, significantly enhancing its performance in the open-set classification of HIS data. To understand the impact of these two components, we conducted ablation experiments by removing these modules from FIWPPN and then performing open-set classification on the "Unknown" labeled data across the three datasets, using open-set classification accuracy as the evaluation metric. The results are presented in Table VIII.

According to Table VIII, the integration of domain-invariant features and the weighted Pearson distance similarity measure is crucial to FIWPPN's success in open-set classification. Removing either the invariant feature module or the weighted Pearson module from FIWPPN led to a considerable decline in open-set classification accuracy.

Specifically, without the invariant feature module, the model experienced a decrease of 22.32%, 23.78%, 22.97%, 22.57%, and 20.7% in average open-set classification accuracy for training datasets of 20, 50, 100, 150, and 200, respectively. With the removal of the weighted Pearson distance metric, the average open-set classification accuracy dropped by 27.11%, 26.89%, 25.62%, 24.23%, and 22.95%. The domain-invariant feature applies linear interpolation to mix labels of different samples, thereby expanding the training data's distribution and enhancing the model's generalization capability. The weighted Pearson distance metric evaluates the similarity of test data in terms of the metric domain, thus enabling the detection and classification of data with unknown labels.

In conclusion, the FIWPPN model, employing domain-invariant features and a weighted Pearson distance similarity measure, excels in the HIS open-set classification. Compared to other algorithms, FIWPPN stands out in unknown class identification, model stability, classification effectiveness, and convergence speed. However, despite its effective identification of open sets of known and unknown classes in HIS data following boundary enhancement, some stray data points in the sample space remain unreasonably classified. Moreover, the model's reliance on a thresholding method for open-set recognition somewhat impacts its accuracy. These limitations necessitate further refinement in future research.

V. Conclusion

Addressing the challenges of open-set classification in HSI within remote sensing technology, this article introduces a FIWPPN. We thoroughly detail the construction and implementation of this prototype network. Experimental results demonstrate that FIWPPN significantly excels in managing HSI open-set classification, surpassing other advanced classification methods.

The research presented in this article offers novel insights and methodologies for HSI open-set classification, bearing substantial theoretical and practical relevance for the advancement of remote sensing science. Future studies could delve into refining the FIWPPN model, aiming to further amplify feature invariance to enhance boundary delineation across diverse HSI classification tasks. In addition, the application of the FIWPPN method in open-set classification challenges across various domains could broaden its utility in different fields.

However, our work does encounter challenges, such as stray data points and thresholding methods that impact classification accuracy. Our future research endeavors will concentrate on these areas to develop more proficient open-set classification models for HIS data. Concurrently, we aim to expand our exploration into open-set classification models that can accommodate new classes not originally part of the HSI images.

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